

Boundary Case: Minimal Governance Capacity

What Minimum Viable Human Governance Looks Like for a CKS Deployment, Which Governance Activities Are Mandatory vs. Deferrable Under Capacity Constraints, and Where Reduction Below Minimum Tips From Boundary Case Into Anti-Pattern

Note B6.09 — CKS Derivation Note Series

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Abstract

The CKS architecture requires human governance as an architectural commitment: humans retain the rights to inspect, modify, and override substrate content and orchestration rules at all times. The architecture does not specify minimum staffing levels. This boundary case note examines a deployment operated by one or two governance practitioners covering all governance activities — the minimum viable human governance configuration. The note identifies which governance activities are mandatory and cannot be deferred without violating architectural commitments, which activities are deferrable or batchable under capacity constraints, and how the labor allocation framework reduces minimum viable capacity requirements while preserving human decision authority. The minimum viable governance boundary is characterized by three conditions: all mandatory-trigger events receive a genuine human governance response; no architectural commitment is structurally absent from the deployment design; and governance decisions are genuinely human-made rather than delegated to automated assessment. Below this boundary, three anti-patterns develop — Governance Theater, entity-level governance neglect, and LLM Governance Conflation. The note concludes that minimal governance capacity is a viable deployment configuration, not an architectural failure, provided the minimum is accurately characterized and maintained.

1. Configuration Description

The deployment under examination is operated by one or two human governance practitioners who bear collective responsibility for all governance activities across the deployment. The governance architecture is complete: all CKS commitments apply without modification. What is constrained is not the architecture but the capacity available to execute governance activities against that architecture.

The practitioners must cover mutation event governance — reviewing and authorizing LLM updates before deployment; entity lifecycle governance — authorizing cell births and deciding on critical cell deaths; conflict registry attention — resolving high-impact conflicts that surface through the A1.03 mechanism; action-feedback Stage 2 review — reviewing queued proposals generated by Stage 1 automated summarization; ongoing governance tests — periodically verifying that the deployment

continues to satisfy its governance commitments; directed selection review — authorizing proposals to deliberately evolve substrate content toward specified objectives; and horizontal evolution peer review — reviewing structurally significant substrate changes.

At full staffing, these activities can be addressed as they arise without prioritization pressure. At minimal capacity, they compete for the same one or two practitioners' time. The boundary case is whether minimal capacity is sufficient to maintain the architecture's governance commitments, or whether it forces compromises that convert a boundary case into an anti-pattern.

2. Architectural Boundaries Tested

Three architectural commitments are under pressure in this configuration.

A1.01 minimum viable human governance. The architecture requires human governance but does not specify minimum staffing. The question this boundary case surfaces is what minimum governance presence is required for A1.01 to hold in practice. The three governance rights — inspect, modify, override — do not require full staffing to be architecturally genuine. A single practitioner who retains and exercises those rights maintains A1.01. What minimal capacity tests is whether a single or pair of practitioners can cover the trigger events those rights must respond to without the rights becoming nominal rather than exercised.

Two-stage action-feedback at minimal capacity. The action-feedback mechanism involves a Stage 1 automated summary and a Stage 2 human review before any proposed substrate modification takes effect. Stage 2 requires human attention. At minimal capacity, the cadence for Stage 2 review may slow substantially — proposals queue until review capacity becomes available. The boundary question is whether a slowed Stage 2 cadence violates the architecture or merely delays it. The architecture requires that Stage 2 occur before proposals take effect; it does not require that Stage 2 occur within any particular time window. Proposal queuing is therefore architecturally permissible; Stage 2 bypassing is not.

Operational timescale governance readiness. Governance readiness at operational timescales requires that governance capacity be available to respond to time-sensitive events — mutation events, lifecycle requests, high-impact conflicts — within operationally acceptable windows. Minimal capacity may not sustain continuous availability. The boundary question is whether governance response latency under minimal capacity remains within acceptable operational bounds, or whether it becomes a structural gap.

3. Governance Triage: Mandatory vs. Deferrable

The central practical contribution of this boundary case is the identification of governance triage — the explicit distinction between activities that must be addressed when they arise and activities that can be batched, delayed, or addressed at cadence windows without violating any architectural commitment.

3.1 Mandatory Activities

Mandatory governance activities are those where deferral would leave an architectural commitment structurally unmet — either an event that requires governance authorization would proceed without it, or a commitment that requires continuous validity would lapse.

Mutation event governance. LLM updates require governance verification before deployment. This commitment — that instinct-layer changes be reviewed and authorized before they take effect in the deployment — admits no deferral. An LLM update that deploys without governance review violates the instinct evolution governance commitment regardless of how constrained governance capacity is. Governance must respond to mutation events as they occur, even if response latency under minimal capacity is longer than at full staffing. The mandatory character of this activity means minimal-capacity deployments should avoid deploying LLM updates at high frequency unless they have structured the verification process to be executable within available capacity windows.

Entity lifecycle governance. Cell births require governance authorization. Critical cell deaths — those where archival decisions, lineage implications, or operational consequences are significant — require governance decision. These lifecycle events cannot proceed as ungoverned operations. A cell birth that occurs without authorization is structurally equivalent to the cell-birth-without-governance anti-pattern regardless of whether governance was merely unavailable or deliberately bypassed. Under minimal capacity, lifecycle governance may impose queue behavior — birth requests wait for the next governance review window — but authorization must occur before birth completes.

High-impact conflict registry attention. The conflict registry records first-class substrate conflicts per the A1.03 commitment. Most conflicts can remain in the registry without immediate resolution; that is the registry's purpose. But a subset of conflicts have operational implications that require timely governance attention — conflicts whose unresolved state blocks downstream substrate operations, conflicts flagged as requiring cross-cell coordination decisions, or conflicts involving governance-sensitive content. These high-impact conflicts constitute mandatory governance triggers. Minimal-capacity practitioners must identify which registry entries meet the high-impact threshold and address them within operationally acceptable windows, while lower-impact registry entries remain queued for batch review.

3.2 Deferrable and Batchable Activities

Deferrable governance activities are those that can be addressed at review windows, in batches, or at reduced frequency without leaving any architectural commitment unmet during the interval between reviews.

Action-feedback Stage 2 review. Stage 1 summarization of action-feedback signals can proceed continuously as an automated governance-labor activity. Proposals generated by Stage 1 queue until Stage 2 human review capacity becomes available. The architecture requires Stage 2 review before any proposal takes effect; it does not require that the queue be cleared continuously. A minimal-capacity deployment may address Stage 2 review at scheduled windows — weekly, or at whatever cadence governance capacity allows — with proposals queued in the interim. The queue mechanism is not an architectural compromise; it is how the architecture accommodates variable review cadence without allowing proposals to take effect without human decision.

Ongoing governance test suite. Ongoing governance tests verify that the deployment continues to satisfy its commitments. These tests can occur at reduced frequency under minimal capacity, provided frequency remains sufficient to detect drift before it becomes entrenched. The tests are architectural monitoring, not architectural enforcement; their reduction in frequency under capacity pressure is a

deferral of monitoring, not a suspension of commitment.

Horizontal evolution peer review. Reviews of structurally significant substrate changes can be batched at governance review windows. Individual structural changes do not require immediate review provided they do not take effect until review occurs. Batching these reviews at scheduled windows reduces the number of distinct review events governance must attend to, without compromising the review requirement itself.

3.3 Labor Reduction Through A1.12

The labor allocation framework distinguishes governance authority from governance labor. Labor can be allocated to LLMs operating under human direction; authority cannot. Under minimal capacity, this distinction is the primary mechanism for making minimum viable governance achievable.

LLMs can draft directed selection proposals for governance authorization — reducing the practitioner's task from authoring to reviewing and deciding. LLMs can execute automated verification gates for mutation events — running structured verification sequences and producing summary records for governance review, reducing the practitioner's task from executing verification to reviewing results and authorizing deployment. LLMs can generate structured summaries of the conflict registry — flagging high-impact conflicts, categorizing pending entries by type and significance — reducing the practitioner's task from surveying the full registry to reviewing a governance-oriented summary. LLMs can execute Stage 1 action-feedback summarization, queuing structured proposals that capture the operationally generated signal into governance-review-ready form.

In each case, LLM labor reduces the practitioner time required for mandatory governance activities without transferring any governance decision to the LLM. The practitioner reviews results and makes decisions; the LLM performs the labor of preparation. This is the A1.12 architecture in its most operationally significant application: minimal capacity is achievable precisely because governance authority and governance labor are separate commitments.

4. Minimum Viable Governance Characterized

Minimum viable governance for a CKS deployment is characterized by three conditions that must hold simultaneously.

Condition 1: All mandatory-trigger events receive a genuine human governance response. Every mutation event produces a governance authorization before deployment. Every entity lifecycle request produces a governance decision before completion. Every high-impact conflict registry entry receives governance attention within an operationally acceptable window. These responses may be slow, LLM-labor-assisted, and batched where the trigger type permits batching. They must occur. A deployment where any category of mandatory trigger events routinely proceeds without governance response has crossed from minimal capacity into structural absence of governance for that trigger type.

Condition 2: No commitment is structurally absent from the deployment design. All architectural commitments apply and are designed-for in the deployment's governance structure. The action-feedback mechanism exists and queues proposals for Stage 2 review — even if Stage 2 review cadence is slow. The conflict registry exists and records conflicts — even if batch resolution cadence is infrequent. The

governance test suite exists and runs — even if less frequently than at full staffing. A commitment that has been removed from the deployment design because of capacity pressure is not a deferred commitment; it is an absent commitment, and its absence is an architectural failure.

Condition 3: Governance decisions are genuinely human-made. The practitioner reviews and decides; LLM-generated summaries, proposals, and verification results inform the decision without replacing it. This condition is operationally tested by asking whether the practitioner could override any specific output of the LLM-labor process — whether a verification gate result, a conflict registry summary flag, or a Stage 1 proposal could be rejected or modified by the practitioner on the basis of their own review. If yes, the decision is genuinely human-made and LLM labor is functioning as designed. If the practitioner's role has effectively reduced to rubber-stamping LLM outputs because there is insufficient capacity for genuine review, the governance is theater rather than genuine governance.

5. Boundary Tests

Four boundary tests determine whether a minimal-capacity deployment is operating within the minimum viable boundary or has crossed into anti-pattern territory.

(a) Are all mandatory governance activities being executed per their triggers? The test is whether mutation events, lifecycle requests, and high-impact conflicts each produce a genuine governance response before the corresponding action takes effect. The test is not whether governance responds within a specific time window; it is whether governance responds at all before the trigger-event completes. A deployment that logs trigger events without governance response is not a slow deployment — it is an ungoverned one for that trigger type.

(b) Are deferrable activities being addressed within acceptable cadence? The test is whether Stage 2 proposal queues are reviewed, governance tests are run, and horizontal evolution reviews occur at a frequency that prevents commitment drift from becoming entrenched. There is no architecturally specified cadence for deferrable activities; the test is whether the cadence chosen by governance practitioners is sufficient to detect and address drift before it compounds.

(c) Is governance triage explicit? The test is whether governance practitioners know which activities are mandatory and which are deferrable. Implicit triage — where practitioners default to addressing whichever activities are most salient rather than which are architecturally mandatory — is a governance theater risk. Explicit triage means the practitioners have a written or operationally stable understanding of which events must trigger governance response regardless of queue pressure, and which events can wait.

(d) Is LLM labor used to reduce governance burden while maintaining human decision authority? The test is whether practitioners can identify, for any LLM-generated summary or recommendation, what decision they are making and on what basis. LLM labor that is invisible to the practitioner — where the practitioner cannot reconstruct what the LLM produced and why the practitioner accepted or modified it — is not labor reduction under human direction. It is authority transfer, which is structurally identical to LLM Governance Conflation regardless of intent.

6. Stress Points

Three stress points are particularly acute under minimal capacity.

Governance Theater risk (B3.22 analog). Under capacity pressure, the temptation is to compress review to maintain throughput — to approve proposals, authorize lifecycle events, and pass mutation verification results faster than genuine review would permit. The result is governance theater: the formal structure of governance is present, but the substance of genuine review and decision is absent. The minimal-capacity configuration makes this stress point more acute than any other configuration because capacity pressure is persistent rather than episodic. Governance practitioners at minimal capacity must actively protect the substance of mandatory review even when review capacity is limited; the architecture requires that decisions be genuinely made, not that they be made quickly.

Entity-level governance neglect. Under capacity pressure, governance attention may concentrate on deployment-level activities — high-visibility events, major substrate changes, system-wide evolution decisions — while entity-level governance becomes perfunctory. Individual cell births may receive cursory authorization; cell death decisions may be deferred past reasonable operational windows; per-cell conflict registry attention may lapse. This produces a deployment where deployment-level governance is genuine but entity-level governance is structurally thin. The cell is the architectural unit of the CKS pattern; deployment-level governance that neglects entity-level governance is not full governance of the deployment.

LLM Governance Conflation risk. As LLM labor reduces the visible effort of mandatory governance activities, the risk increases that practitioners treat LLM-generated assessments — verification gate passes, conflict severity classifications, proposal quality evaluations — as governance decisions rather than as governance inputs. The distinction is not administrative; it is architectural. LLM-generated assessments are substrate mediator outputs operating under human direction; governance decisions are human judgments that take those outputs as inputs. When the distinction collapses — when practitioners sign off on LLM assessments rather than making decisions informed by LLM-generated material — the architecture's human governance commitment has failed even though the formal process may look intact.

7. Architectural Limits: The Minimum Viable Boundary

The minimum viable governance boundary is not a staffing count. It is a functional characterization: the deployment configuration below which the architecture's governance commitments become nominal rather than genuine.

The architecture's minimum is where all three conditions in §4 hold simultaneously: mandatory triggers receive human governance response, no commitment is structurally absent, and governance decisions are genuinely human-made. A deployment that meets all three conditions is human-governed in the CKS sense regardless of how small its governance team is. A deployment that fails any one of the three conditions is not at the minimum — it is below the minimum, and the specific condition it fails identifies the anti-pattern it has entered.

Failure of Condition 1 — mandatory triggers proceeding without human governance response — is structurally equivalent to the cell-birth-without-governance or instinct-evolution-without-verification anti-patterns applied to whichever trigger type is unresponded to.

Failure of Condition 2 — a commitment structurally absent from the deployment design — is an architectural incompleteness that staffing cannot explain. A deployment that removes the action-feedback Stage 2 mechanism because governance capacity is too thin to operate it has not adapted the architecture to its capacity constraints; it has violated the architecture.

Failure of Condition 3 — governance decisions not genuinely human-made — is Governance Theater, the anti-pattern where governance form is preserved while governance substance is lost.

The practical implication is that minimal-capacity operators must explicitly audit their deployment against all three conditions before declaring minimal capacity operational. The audit should identify which mandatory triggers are covered, which deferrable activities have established cadences, and which LLM labor outputs are reviewed against independent governance judgment rather than accepted on the basis of the LLM's own assessment. A deployment that passes this audit is operating at minimum viable governance. A deployment that has not conducted this audit may be operating below minimum without the operators' awareness.

8. Relationship to Adjacent Notes

This note characterizes the minimum viable boundary from the inside — the conditions that define minimum viable governance and the stress points that push deployments below that boundary under capacity pressure. The adjacent boundary cases approach the governance architecture from different pressure vectors. B6.08 examines the specific challenge of instinct evolution events occurring during active cell execution — a trigger type that is mandatory under this note's characterization and that poses distinctive coordination challenges at minimal capacity. B6.10 will examine large-scale cell populations, where the minimum viable governance boundary applies to a substrate that may generate mandatory triggers at higher absolute frequency than a small deployment, testing whether per-trigger governance response remains tractable as cell population scales.

The anti-pattern notes B3.21, B3.22, and B3.25 formalize the failure modes this note identifies as stress points. This note's contribution relative to those anti-pattern notes is the positive characterization: rather than naming only the failure modes, it characterizes the minimum viable governance configuration that avoids them under constrained capacity.

This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

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